

CHE 565 -- Homework 3

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Problem 1

Setup

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At $x_0 = [1, 1]$, $f(x_0) = 0.5$

Gradient of f :

$$\begin{bmatrix} 1.0x_1 + 1.0x_2 - 1 \\ 1.0x_1 + 2.0x_2 - 1 \end{bmatrix}$$

At $x_0 = [1, 1]$, $Df(x_0) = \begin{bmatrix} 1. \\ 2. \end{bmatrix}$

Function $im(g(\alpha) = x - \alpha * Df$:

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 - 1.0\alpha \\ 1 - 2.0\alpha \end{bmatrix}$$

Function ϕ_k :

$$\phi(\alpha) = 2.0\alpha^4 - 6.0\alpha^3 + 8.5\alpha^2 - 5.0\alpha + 0.5$$

Derivative of ϕ_k :

$$\frac{d}{d\alpha}\phi(\alpha) = 8.0\alpha^3 - 18.0\alpha^2 + 17.0\alpha - 5.0$$

Optimal α : $[0.5000000000000000, 0.875 - 0.695970545353753*I, 0.875 + 0.695970545353753*I]$